

Lab Sheet 03

Part 01 : Probability Distributions

Sample Codes for Binomial Distribution in R

1. `dbinom()` - Probability Mass Function (PMF)

Calculates the probability of exactly x successes in n trials with success probability p .

`dbinom(x, size, prob)`

- **x** – Number of successes
- **size** – Number of trials
- **prob** – Probability of success in each trial

Example:

Probability of exactly 3 successes in 10 trials with $p = 0.5$

`dbinom(3, size = 10, prob = 0.5)`

2. `pbinom()` - Cumulative Distribution Function (CDF)

Calculates the cumulative probability up to x successes.

`pbinom(q, size, prob, lower.tail = TRUE)`

- **q** – Number of successes
- **size** – Number of trials
- **prob** – Probability of success
- **lower.tail** – TRUE for $P(X \leq x)$, FALSE for $P(X > x)$

Example:

Probability of 3 or fewer successes in 10 trials with $p = 0.5$

`pbinom(3, size = 10, prob = 0.5)`

Probability of more than 3 successes

`pbinom(3, size = 10, prob = 0.5, lower.tail = FALSE)`

3. `qbinom()` - Quantile Function (Inverse CDF)

Finds the number of successes corresponding to a given cumulative probability.

`qbinom(p, size, prob, lower.tail = TRUE)`

- **p** – Probability value (between 0 and 1)
- **size** – Number of trials
- **prob** – Probability of success
- **lower.tail** – TRUE for lower quantile, FALSE for upper quantile

Example:

Number of successes corresponding to 0.5 cumulative probability

`qbinom(0.5, size = 10, prob = 0.5)`

4. `rbinom()` - Random Sampling

Generates random samples from a binomial distribution.

`rbinom(n, size, prob)`

- **n** – Number of random samples
- **size** – Number of trials
- **prob** – Probability of success

Example:

Generate 10 random values from a binomial distribution

`rbinom(10, size = 10, prob = 0.5)`

1. Binomial Distribution

Question 01) A factory produces light bulbs, and the probability of a bulb being defective is 0.1. Out of a batch of 10 bulbs, what is the probability that

1. Exactly 2 bulbs are defective
2. At most 2 bulbs are defective

$n = 10$

$p = 0.1$

$\text{prob_exactly_2} = \text{dbinom}(2, \text{size} = n, \text{prob} = p)$

$\text{prob_at_most_2} = \text{pbinom}(2, \text{size} = n, \text{prob} = p)$

$\text{print}(\text{prob_exactly_2})$

$\text{print}(\text{prob_at_most_2})$

Question 02) A marketing survey finds that 40% of customers prefer brand A. In a random sample of 15 customers, what is the probability that

1. Exactly 6 customers prefer brand A
2. More than 6 customers prefer brand A

2. Hypergeometric Distribution

Question 01) A deck of cards has 52 cards, including 13 hearts. If 5 cards are drawn without replacement, what is the probability of getting exactly 2 hearts

$N = 52$

$K = 13$

$n = 5$

```
prob_2_hearts = dhyper(2, K, N - K, n)
```

```
print(prob_2_hearts)
```

Question 02) In a batch of 20 items, 7 are defective. If 6 items are randomly chosen, what is the probability that

1. Exactly 3 items are defective
 2. At least 3 items are defective
-

3. Poisson Distribution

Question 01) A call center receives an average of 5 calls per minute. What is the probability of receiving exactly 7 calls in a minute

```
lambda = 5
```

```
prob_7_calls = dpois(7, lambda)
```

```
print(prob_7_calls)
```

Question 02) If a website gets an average of 10 hits per second, what is the probability that

1. No hits are received in a second
 2. At least 5 hits are received in a second
-

4. Geometric Distribution

Question 01) A die is rolled until a 6 appears. What is the probability that it takes exactly 4 rolls to get a 6

```
p = 1/6
```

```
prob_4_rolls = dgeom(3, prob = p)
```

```
print(prob_4_rolls)
```

Question 02) The probability of a customer making a purchase at a store is 0.2. What is the probability that the first purchase occurs on the 5th customer

5. Negative Binomial Distribution

Question 01) A basketball player has a 60% chance of making a free throw. What is the probability that the player makes the 5th successful shot on the 8th attempt

`r = 5`

`p = 0.6`

`prob_5th_success = dnbinom(3, size = r, prob = p)`

`print(prob_5th_success)`

Question 02) A factory produces widgets, and 10% of them are defective. What is the probability that the 3rd defective widget is found on the 7th inspection

6. Exponential Distribution

Question 01) The average time between arrivals of buses at a station is 10 minutes. What is the probability that the next bus arrives within 5 minutes

`rate = 1/10`

`prob_within_5 = pexp(5, rate = rate)`

`print(prob_within_5)`

Question 02) A machine part has an average lifespan of 100 hours. What is the probability that it fails after 120 hours

7. Normal Distribution

Question 01) The heights of students in a class are normally distributed with a mean of 170 cm and a standard deviation of 10 cm. What is the probability that a randomly selected student is taller than 180 cm

mean = 170

sd = 10

prob_taller_180 = pnorm(180, mean, sd, lower.tail = FALSE)

print(prob_taller_180)

Question 02) IQ scores are normally distributed with a mean of 100 and a standard deviation of 15. What is the probability that a person's IQ is between 85 and 115

Question 03) A company's weekly sales are normally distributed with a mean of \$50,000 and a standard deviation of \$5,000. What is the probability that the sales are less than \$40,000

Question 04) What is the z-score for a value of \$60,000 in the above case

Question 05) Given the above distribution, find the 95th percentile of weekly sales

8. Gamma Distribution

Question 01) The time to complete a task follows a Gamma distribution with a shape parameter of 2 and a scale parameter of 3. What is the probability of completing the task in less than 5 units of time

shape = 2

scale = 3

prob_less_5 = pgamma(5, shape = shape, scale = scale)

print(prob_less_5)

Question 02) If the same task follows the above Gamma distribution, what is the density value at 4 units of time

9. Beta Distribution

Question 01) The proportion of defective items in a batch follows a Beta distribution with shape parameters $\alpha = 2$ and $\beta = 5$. What is the probability that the proportion is less than 0.3

alpha = 2

beta = 5

```
prob_less_0_3 = pbeta(0.3, shape1 = alpha, shape2 = beta)
```

```
print(prob_less_0_3)
```

Question 02) What is the mode of the Beta distribution in Question 1

Part 02 : Data Cleaning in R

Data cleaning is an essential step in data analysis, ensuring your dataset is accurate, consistent, and suitable for analysis. This lab sheet covers three key aspects of data cleaning in R

- Handling Missing Values
- Handling Duplicates
- Handling Outliers

We will use manually created data frames to demonstrate these concepts.

1. Handling Missing Values

Sample Dataset with Missing Values

```
df = data.frame(  
  ID = 1:8,  
  Name = c("Alice", "Bob", "Charlie", "David", NA, "Frank", "George", "Helen"),  
  Age = c(25, NA, 30, 45, 50, NA, 34, 29),  
  Score = c(85, 90, NA, 88, 75, 80, 95, NA)  
)  
print(df)
```

Detecting Missing Values

```
# Check for NA values
```

```
table(is.na(df))
```

```
# Column-wise count of missing values
```

```
colSums(is.na(df))
```

Handling Missing Values

Method 1: Removing Rows with Missing Values

```
df_clean = na.omit(df)
```

```
print(df_clean)
```

Method 2: Filling Missing Values with Mean/Median/Mode

```
# Fill Age column with mean
```

```
df$Age[is.na(df$Age)] = mean(df$Age, na.rm = TRUE)
```

```
# Fill Score column with median
```

```
df$Score[is.na(df$Score)] = median(df$Score, na.rm = TRUE)
```

```
print(df)
```

Method 3: Using mice Library for Advanced Imputation

```
library(mice)
```

```
df_imputed = mice(df, method = "pmm", m = 1)
```

```
df_cleaned = complete(df_imputed)
```

```
print(df_cleaned)
```

2. Handling Duplicates

Sample Dataset with Duplicates

```
df_duplicates = data.frame(
```

```
  ID = c(1, 2, 3, 3, 4, 5, 5, 6),
```

```
  Name = c("Alice", "Bob", "Charlie", "Charlie", "David", "Frank", "Frank",  
"George"),
```

```
  Age = c(25, 30, 30, 30, 40, 50, 50, 60)
```

```
)
```

```
print(df_duplicates)
```

Detecting Duplicates

```
# Identify duplicates
```

```
duplicated(df_duplicates)
```

```
# Display duplicate rows
```

```
df_duplicates[duplicated(df_duplicates), ]
```

Removing Duplicates

Method 1: Remove All Duplicate Rows

```
df_unique = df_duplicates[!duplicated(df_duplicates), ]  
print(df_unique)
```

Method 2: Keep Only the Last Occurrence

```
df_last = df_duplicates[!duplicated(df_duplicates, fromLast = TRUE), ]  
print(df_last)
```

3. Handling Outliers

Sample Dataset with Outliers

```
df_outliers = data.frame(  
  Student = c("A", "B", "C", "D", "E", "F", "G"),  
  Scores = c(78, 85, 90, 95, 150, 200, 88) # 150 and 200 are outliers  
)  
print(df_outliers)
```

Detecting Outliers

Method 1: Using Boxplot

```
boxplot(df_outliers$Scores, main = "Boxplot for Scores")
```

Method 2: Using IQR (Interquartile Range)

```
Q1 = quantile(df_outliers$Scores, 0.25)
```

```
Q3 = quantile(df_outliers$Scores, 0.75)
```

```
IQR_value = Q3 - Q1
```

Identifying Outliers

```
outlier_condition = df_outliers$Scores < (Q1 - 1.5 * IQR_value) |  
df_outliers$Scores > (Q3 + 1.5 * IQR_value)  
  
print(df_outliers[outlier_condition, ])
```

Handling Outliers

Method 1: Removing Outliers

```
df_no_outliers = df_outliers[!outlier_condition, ]  
  
print(df_no_outliers)
```

Method 2: Capping Outliers (Winsorizing)

Cap extreme values at upper and lower thresholds

```
df_outliers$Scores[df_outliers$Scores > (Q3 + 1.5 * IQR_value)] = Q3 + 1.5 *  
IQR_value  
  
df_outliers$Scores[df_outliers$Scores < (Q1 - 1.5 * IQR_value)] = Q1 - 1.5 *  
IQR_value  
  
print(df_outliers)
```